

TromboneAlign

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Executive Summary

The TromboneAlign project addressed a common yet overlooked challenge faced by student trombone players: physical strain and discomfort during extended practice sessions. After conducting user interviews, research, and analysis of existing market products, the team found that many trombone players experience pain, particularly in the wrist and shoulder, due to incorrect instrument grip technique and weight distribution. The team's goal was to design a simple, ergonomic grip that supports proper hand positioning, reduces strain, and encourages healthy long-term playing habits.

The final product, the TromboneAlign, is a lightweight, adjustable trombone grip that clips onto the trombone's horizontal slide. The grip includes two finger holes to guide correct hand placement while allowing the slide to move freely. It's removable, accommodates a range of hand sizes, and can be installed in under one minute. The prototype was developed through iterative brainstorming, CAD modeling, and material testing. Verification testing confirmed the grip meets design standards for durability, weight, ease of use, and comfort.

The TromboneAlign supports better physical ergonomics, thus reducing pain, promoting proper technique, and resulting in longer, more and less painful practice sessions. It has value for a wide range of stakeholders—students, teachers, music retailers, and parents—as it is an affordable and effective solution to player strain. With the product's user group validating its benefits, the TromboneAlign has strong potential for market adoption in both music education and music recreation.

Problem Definition Review

1. Introduction

Each member of the team interviewed someone they knew to uncover potential user pains in a wide range of user groups and life domains. Eventually, the discussion reached the topic of musicians. Everyone in the team had played an instrument and experienced user pains, but after a long discussion, the team mutually agreed to focus on the user pains of a Trombone Player, as they had unresolved issues the team could cater to.

This design project was focused on resolving user pains faced by trombone players. Many student musicians struggled to balance their academic responsibilities with artistic hobbies, and they often found regular trombones painstaking to play, heavy, and difficult to transport. In addition, students typically had limited funds which made affordability a significant concern.

There was an opportunity to design a product for trombone players that could resolve their user pains and fulfill their user needs. Thus, the purpose of this document is to show each step the team followed during the process of designing this product. In addition, this document outlines the full engineering design process the team followed to develop the TromboneAlign grip from identifying user pains and analyzing market needs to prototyping, testing, and refining a product that meets the ergonomic and economic needs of student trombone player

2. Problem Definition

Introduction:

To better understand the targeted user group and design a solution that directly addresses their needs, the team developed a detailed user persona of a hypothetical user. This persona represented a typical college trombone player experiencing hand strain and general discomfort that posed an obstacle for long practice sessions. The following figure and description highlight key demographic, behavioral, and ergonomic insights about this user, which informed the team's product design decisions and priorities throughout the project.

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a. User Persona



Figure 2.1: Graphical User Persona

To better understand the daily challenges faced by the hypothetical user, the team identified several routine tasks that the user in question would perform, and the specific pains associated with each. These tasks were chosen based on interview data and user observations. The following list outlines the key discomforts that commonly affected trombone players during practice and performance.

b. Tasks and Pains

1.) Practicing for an extended period

- Lips get tired of supporting airflow.
- Arm gets tired of moving the slide.
- The heavy trombone becomes difficult to hold as fatigue sets in.

2.) Flipping a page

- The musician may play a wrong note if distracted by flipping a page.
- The book could fall off the stand if flipped too quickly.
- Outdoor wind can blow the pages away.

3.) Draining Spit

- Buildup of spit in the instrument while playing can affect the sound of the instrument.
- If indoors, players cannot just drain spit onto the floor.
- Hygiene concerns due to spit buildup.

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3. Research Plan

The team intended to start research by learning more about the trombone players themselves since most team members lacked insight into what it was like to play the trombone. After that, the focus of the research was shifted more towards the physical pains trombone players faced so the team could figure out what specific issue to design for and market towards. Research data about physical injuries from trombone players was searched for. The task of collecting this research was divided among the team. Many of the sources uncovered from this research came from official journals or were anecdotes by experienced players. Research on trombone players and their physical injuries was required to meet ethical standards, including informed consent, privacy, and bias control. Participants' information needed to be kept confidential, and their experiences respected.

Table D.1: Research Plan Table

Research Question	Quantitative Data	Qualitative Data
1.) Why do you want to play an instrument?	The Child as Musician: A Handbook of Musical Development [1]	Why Play – Trombone [2]

2.) What are the psychological benefits of playing the trombone?	A Prescription for Music Lessons [3]	The Psychological Benefits of Playing Brass Instruments: More Than Just Music [4]
3.) How does playing the trombone influence students' academic performance?	Investigating trainee music teachers' beliefs on musical abilities and learning: a quantitative study [5]	Relations among Motivation, Performance Achievement, and Music Experience Variables in Secondary Instrumental Music Students [6]
4.) How much time do trombone players typically practice in week?	If You Practice [7]	Trombone Fundamentals: How to Develop Your Own Daily Routine [8]
5.) What physical pain do trombone players have from playing?	Musculoskeletal and Psychological Assessments used in quantitatively based studies about musicians' health in brass players: A systematic literature review [9]	Play Without Pain: Repetitive Use Injuries and What to Do About Them [10]

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Based on the results of the research, the team compiled the most evidential tasks and pains experienced by trombone players that could have informed the design process. In this section, the team focused on the task of practicing for an extended period, which emerged as a significant cause of physical strain. The following pain points were drawn directly from the team's research and provided a foundation for understanding the user group's needs and guiding the product development.

4. Research Results

This phase of the project focused on identifying and understanding the challenges that were commonly experienced by student trombone players. Through user interviews and research, the team observed that long practice sessions consistently emerged as a primary source of physical strain. As a result, long practice sessions were chosen as a focal point for the project, as it directly contributed to strain to the wrist, shoulder, and arm pain that

many trombone players face.

a. Creating User Value

Task: Long practice sessions.

i. Pains

- 1.) The constant pressure in the lips needed to maintain consistent airflow causes pain and mouth fatigue.
- 2.) The arm gets tired of moving the slide, and the shoulder gets tired of being held in place.
- 3.) The trombone feels heavier overtime as the player cannot set it down.
- 4.) The greater likelihood of experiencing playing related musculoskeletal disorders than the players of any other brass instrument. [9]

ii. Gains

- 1.) Improvement in fine muscle coordination and efficient neural pathways for learned music. [7]
- 2.) Enhanced memory, concentration, and multi-tasking skills. [1]
- 3.) Improved creativity through music, inspiring the player to innovate in other fields. [1]
- 4.) Improvement in everyday posture when learning the correct posture to play the trombone. [22]

iii. User Needs

- 1.) Instrument variations or implements that can be used to reduce muscular strain when playing the trombone for prolonged periods.
- 2.) Lighter weight and **easier to carry** accessories for trombone players.

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- 3.) Features that encourage and motivate student players to practice regularly and **engage** with their music while playing.
- 4.) Professional level instrument and mouthpiece, which can **hold the instrument's notes**, giving the player time to think about their next move while holding out a note.
- 5.) Less dense parts or materials to be used in the construction of a trombone to reduce its weight, possibly in a composite with brass.
- 6.) Affordability. The team must ask themselves if an improvement is worth it if it drastically raises costs, and if there may be a lower cost solution to accomplish the same thing or close to the same thing. This is a form of cost-benefit analysis.

iv. Pairwise Comparison Chart

Table D.2: Pairwise Comparison

	Easier to hold instruments	Easier to transport instruments	Encouraging students to practice	Improving quality of instrument's tone	Better material parts	Affordability	Total	Normalized	Predicted
Easier to hold instruments		1	1	1	1	1	5	5.0	5
Easier to transport instruments	0		1	1	1	1	4	4.0	4
Encouraging students to practice	0	0		0	0	0	0	-	1
Improving quality of instrument's tone	0	0	1		0	0	1	1.0	0
Better material parts	0	0	1	1		1	3	3.0	2
Affordability	0	0	1	1	0		2	2.0	3

The team identified the most important user need as making the trombone easier to hold, which received the highest score in the pairwise comparison chart. Holding the instrument requires the left hand to remain in an awkward position for extended periods, often leading to pain in the shoulder, arm, and hand—particularly among student musicians who are still developing proper technique [9][11]. Addressing this issue was prioritized due to its direct impact on players' comfort and long-term health.

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The second highest priority was improving the ease of transporting the instrument. Trombones are bulky and difficult to carry, and finding an appropriate case involves considering several features such as padding, latches, wheels, and overall weight [12]. For student players navigating crowded hallways or public transit, this can be a significant

inconvenience.

The third need was higher-quality materials and components. Instruments made from poor-quality materials tend to lack durability and produce inferior sound. This concern was closely tied to affordability, as professional-grade trombones often cost between \$420 and \$5,000 [13], creating a financial barrier for students. Sound quality, while important, was ranked lower because the team believed that players generally adapt to their instrument's tone as long as it is reliable.

The lowest-ranked user need was encouraging students to practice. While still relevant, it was considered secondary to resolving the physical and financial limitations that directly discourage regular practice. The team reasoned that eliminating these barriers would lead to increased motivation and playing time.

The predicted ranking of users' needs closely aligned with the results of the pairwise comparison chart, reinforcing the team's decision to focus on ergonomic strain and grip support. To further justify this direction, the team analyzed relevant market stakeholders and their potential interactions with the product, as outlined in the following section.

b. Market Character

To ensure the viability of the TromboneAlign grip the team conducted a thorough analysis of the market. This included identifying major stakeholders who would be affected by or benefit from the product, estimating the potential market size, and evaluating current alternatives. This analysis helped the team better understand existing market gaps, assess how the product could address them, and determine the value it could provide beyond student trombone players. The following section outlines the current market landscape to clarify where the TromboAlign fits within it.

v. Stakeholders

1.) Student Trombone Players

- Student trombone players were expected to benefit from a grip that taught them how to hold the trombone properly early in their

development. It would reduce the likelihood of physical pain during the initial stages of learning.

2.) Professional Trombone Players

- Professional trombone players would benefit from improved grip

support that could reduce fatigue and strain during long performances or rehearsals. A more ergonomic grip could help extend their playing career and comfort, especially for those with repetitive strain injuries.

3.) Music Stores

- This product represents a new opportunity for music retailers to expand their accessory offerings. Because the grip is targeted at students and amateur musicians, it could become a widely adopted add-on product, increasing store revenue and customer engagement.

4.) Band Directors

- Band directors could use the grip as a tool to teach correct holding techniques and posture from the start, helping prevent bad habits that lead to injuries. It also helps students play longer and more comfortably, potentially improving performance quality during rehearsals and concerts.

5.) Motivated parents

- Parents of student musicians are likely to be interested in a product that supports their child's physical well-being and encourages continued participation in music. A product that makes playing more comfortable could increase retention and practice time, helping students stick with their musical goals.

vi. Market Size

There was a moderately large market size for this product. Based on rough estimates, approximately 1.9 million individuals in the United States had experience playing the trombone, including current students and those who learned in high school (Figure B.0.1). This number reflected a wide potential user base, ranging from beginners to advanced players. In addition, there were 8,743 musical instrument and supply stores across the U.S. [15], which indicated a strong network of potential distributors and retail partners. These figures suggested that there was a substantial opportunity for product adoption in both the consumer and educational markets, especially if the product was affordable and improved player comfort.

vii. Current Alternatives:

Alternative 1: TromboneSpineSaver™

The TromboneSpineSaver™ holds the trombone, so the player does not

need to, which helps reduce strain on the back, hand, and wrist. It can be adjusted up and down, side to side, and forward and backward. It is clipped on with a belt clip [16]. We expect the team's product will be superior when it comes to allowing players to learn proper technique without the support of a stand, as a stand cannot be used forever.

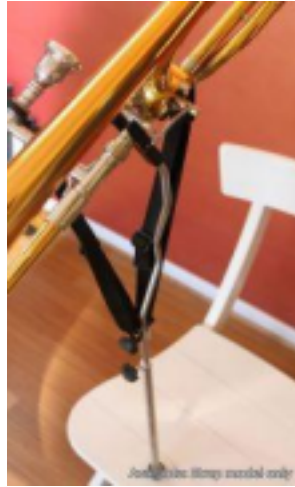


Figure 4.1: Trombone Spine Saver™

Alternative 2: Neotech™ Trombone Ultra-Grip

The Neotech Trombone Ultra Grip supports the trombone's weight, making it feel lighter and easier to hold. It adjusts to fit most trombones and hand sizes, reducing strain and improving control. This allows musicians to focus on playing comfortably with less effort [17]. However, it's also more complicated and less intuitive to use for beginners than the team's design.



Figure 4.2: Neotech™ Trombone Ultra Grip

Alternative 3: Calder Pips Trombone Grip

The Calder Pips Trombone Grip supports the instrument's weight and improves control by reducing strain on the hand. It adjusts to fit different

trombone types and hand shapes, ensuring a comfortable and secure hold. The grips can be reshaped at home for a personalized fit, allowing each player to find the most natural and efficient playing position [18]. Like the previous alternative, however, it is also more complicated and less intuitive than the team's idea.



Figure 4.3: Calder Pips Trombone Grip

Alternative 4: ERGO Brass ERGO Bone

The ERGO bone supports the user's trombone which helps reduce strain and improve posture. It lets the instrument rest on a harness and support rod [19]. This still doesn't address hand pain directly.



Figure 4.4: ERGO Brass ERGO Bone

Alternative 5: Edwards Bullet Brace

The Edwards Bullet brace makes holding the trombone easier by shifting most of the weight to the lower arm muscles. This helps reduce strain on

the wrist and hand. The design is completely adjustable so players can move the brace to a more comfortable position [20]. The team’s product is different in that instead of shifting weight, it improves left hand form, which often resolves support issues more permanently.



Figure 4.5: Edwards Bullet Brace

Table D.3: Partial Competitive Matrix

User Need	Trombone SpineSaver™	Neotech™ Trombone Ultra-Grip	Calder pips Trombone Grip	ERGObrass ERGObone	Edwards Bullet Brace
Easier to hold instruments					
Easier to transport instruments/ accessories					
Encouraging students to practice					
Improving quality of instrument's tone					
Better material parts					
Affordability					

5. Value Proposition

Trombone players, especially students, faced both physical and psychological challenges when practicing for extended periods. One of the most common problems was musculoskeletal pain, particularly in the hands and wrists, caused by awkward grip and the weight of the instrument. There was an indication that these physical strains often discouraged consistent practice, limited performance quality, and increased playing related frustration and anxiety.

This report identified an opportunity to address these pains with a lightweight, ergonomic solution that could improve grip comfort, promote healthier playing habits, and encourage longer, more enjoyable practice sessions. A well-designed grip had the potential to reduce stress, increase musical engagement, and offer student musicians an affordable way to improve both comfort and control while playing.

6. Design Focus

Research revealed that student trombone players frequently experienced physical discomfort, particularly in the wrist and hand, due to the awkward grip and weight distribution of traditional trombones. This physical strain discouraged long practice sessions and may have even contributed to long-term musculoskeletal issues. [9] Affordability and portability were also identified as major concerns, especially for students balancing academic workload with musical commitments.

While several products on the market aimed to improve posture or reduce strain—such as harnesses or wrist supports—they often relied on external hardware, were not widely customizable for student players, or added bulk to the instrument. Additionally, very few existing solutions directly addressed the root cause of poor grip technique, which contributed to discomfort and formation of bad habits over time.

Given these gaps, the design process focused on creating a solution that:

- Made the trombone easier to hold for extended periods
- Encouraged proper grip technique
- Reduced physical fatigue
- It was cost-effective and easy to install

These priorities guided the team's concept development phase and ensured the solution directly addressed the most critical user needs identified through research.

Concept Development Review

7. Concept Brainstorming and Ideation

The first brainstorming sessions focused on identifying the key challenges student trombone players face. One of their biggest struggles was the strain placed on their hands and wrists when holding and playing the instrument for extended periods during long practice sessions. This often led to discomfort and fatigue, which made it difficult for student trombone players to maintain proper playing technique. In early design ideas, the team considered using lighter material to make the trombone but altering the instrument's material could have adversely affected sound quality. Instead of changing the trombone's structure, the focus shifted toward developing an ergonomic grip that enhanced support and didn't interfere with the playing process.

Another important aspect of the grip development project was ensuring the handgrip could adapt to different hand sizes and playing techniques. An initial one-size-fits-all approach proved ineffective, as a single handgrip could not accommodate every hand shape. To make the design more universal, adjustable components allowed players to customize the grip for optimal comfort and support. Additionally, lightweight yet durable materials were selected to provide functional pain relief without adding unnecessary weight to the instrument.

The design fulfilled the user needs defined by the team. The requirements to meet user needs included reducing strain, improving comfort, and the product remaining affordable. The trombone grip offered student musicians a practical solution to many of the previously stated pains, enhancing their overall playing experience. The final design allowed for longer, more comfortable practice sessions while maintaining ease of use and accessibility. This innovative approach showed a commitment to addressing the specific challenges faced by student trombone players, and the team's dedication to creating a product that helped solve their problems.

a. Process Description

The brainstorming process began by researching ergonomic solutions for trombone players. To find potential problems and understand the limitations of the currently existing grips, feedback was collected from potential users. Multiple brainstorming sessions were held, and the team members came up with several ideas for trombone grips. Each idea was evaluated based on how it fits into the user needs, pains and gains, comfort, and producibility.

Brainstorming sessions were focused on solving the most frequent problem that trombone players encounter. Grip pain was determined to be the most frequent issue after looking at the research collected. The team focused on creating an ergonomic grip that could lessen hand and wrist strain in trombone players. Each team member created a design during the brainstorming process to bring the group's vision to reality.

b. Brainstorming Results

During the team's brainstorming sessions, two more ideas came to light: a fixed ergonomic trombone stand to help trombone players maintain stability, and the ability for the trombone grip to have its position adjusted. The adjustable grip concept was the one that was adopted because this idea had potential and flexibility, and it could accommodate many different hand sizes. To test comfort and weight distribution, prototypes and preliminary CAD models were created. Feedback collection from users revealed that a soft surface and adjustable grip would offer a good balance of comfort and usability.

8. Concept Selection

a. Down-Selecting to Two Concepts

Based on the results of the brainstorming session, the two best ideas for further consideration were the fixed ergonomic grip and an adjustable ergonomic grip. The fixed grip would have ensured a tight grip with little adjustability, while players would have been able to adjust the ergonomic grip for their comfort. Adaptability was considered by users to be the most crucial feature because accommodating different hand sizes and playing styles would have set the team's product apart from the current market.

b. Pugh Scoring Matrix

The Pugh scoring matrix states user needs and their importance and compares them to a competitor's design (a trombone stand) and to two of the team's individual designs (Figures B.0.2 and B.0.3). The team decided the most important user need was making it less strenuous to hold the trombone.

Table D.4: Pugh Scoring Matrix

		Reference		Concept 1		Concept 2	
User Need	Weight (1-5)	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score
Easier to Hold the Instrument	5	5	25	5	25	5	25
Easier to transport accessory	4	3	12	3	12	5	20
Encourages students to practice	0	1	0	1	0	1	0
Improves quality of instrument's tone	1	1	1	1	1	1	1
Better Materials	3	1	3	1	3	5	15

Affordability	2	3	6	3	6	5	10
			0		0		0
			0		0		0
			0		0		0
			0		0		0
TOTAL			47		47		71

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c. Final Concept Selection

Before choosing the adjustable ergonomic grip as the final design of the product and prototype, many options were considered. The final concept was selected with trombone players in mind since it provided a unique form of grip support and reduced wrist strain. The adjustability allowed players to modify their grip position, and the use of durable padding ensured the grip was comfortable. It was confirmed by early user comments that this concept solved playing-related issues without harming the sound quality of the trombone. To ensure that was no disruption to the trombone's sound quality and to fine-tune the ergonomic aspects, CAD modeling and prototype testing were used to refine the design.

9. Grand Concept Design

The team designed an ergonomic trombone grip to reduce hand, shoulder, and wrist pain for trombone players. The primary function of the grip was to assist trombone players in maintaining a proper slide grip, which helped alleviate the stress on the wrists and fingers if the wrong technique was being used. This grip was designed to fit the player's natural hand grip. The grip clipped onto the trombone slide, and the player placed their fingers in the holes of the grip. The length of the grip was about 4.0-6.0 inches, and the height was about 1.5-2.0 inches.

While many products existed to help alleviate the pain of trombone players, none encouraged players to change the habits that cause the pain. Based on the team's research, these products didn't focus on the grip technique of the player. Many existing products supported the weight of the trombone with the upper body, rather than the arm. Challenges that were anticipated included not letting the flexibility and durability of the product be suboptimal and making sure the grip stayed fixed to the slide.

In the first preliminary sketch of the trombone grip, it attached partially around the curvature of the slide of the trombone grip (Figure 10.1). Half circles were added to the sides of the grip to act as the "finger holes".

To ensure the product worked well, the team planned to have real trombone players test the grip to determine if it improved comfort and reduced strain. The team also wanted to test whether the grip could be attached to and taken off the trombone slide easily without cracking. If the product proved to be effective and received positive feedback, it could

have received the attention of trombone players, who would likely want to buy it.

10. Prototyping Design

This section outlines the process the team followed to transform the concept of the ergonomic trombone grip into a physical model. In this phase a functional prototype was created and refined to ensure that it met the ergonomic requirements for reducing hand and shoulder pain but that it also met design requirements like ease of use, durability, and comfort for trombone players.

To achieve this, the team established standards by developing design requirements and creating a corresponding verification plan. The design requirements defined key variables that were essential for optimizing the design. The verification plan utilized these

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requirements to create a testing system for evaluating the prototype upon completion. Based on the results, necessary improvements to the prototype were identified and implemented.

Materials, tools, and fabrication methods used to construct the prototype along with the challenges encountered during development and the iterative adjustments made will be documented in the following section. The outcome of this phase was an important prototype.

a. Description of Prototype

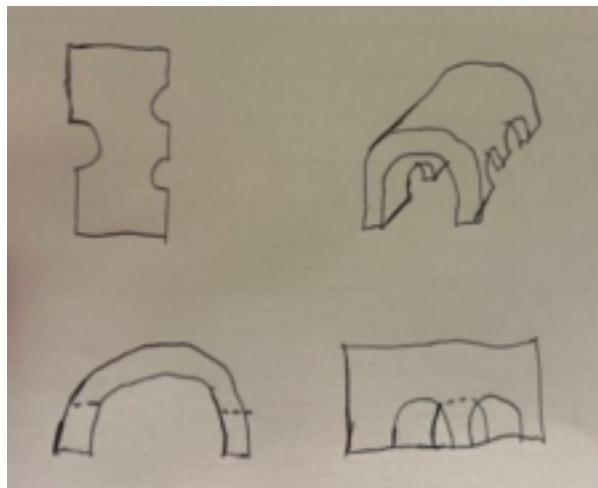


Figure 10.1: Preliminary Sketches of the Ergonomic Trombone Grip

The team made a prototype of the ergonomic trombone grip. The sketch was based on one of the team's brainstorming drawings (Figure A.2). By drawing the initial sketch, the team identified one of the product's limitations, the flexibility of the product. It needed to be strong enough to not break when attached onto the trombone slide, but flexible enough to avoid breaking when the grip was taken off. For the initial model, the team chose to use paper and safety pins to create the hollow semicylinder shape of the product. In the subsequent prototypes, the model was 3D printed with TPU filament.

b. Prototype Design Requirements

The following table outlines the design requirements that the team determined to be important. Each requirement was chosen to avoid potential pitfalls in the product. Goals were established for each design requirement.

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Table D.5: Design Requirements

Design Requirement:	Value(s):
Trombone grip should be lightweight.	Goal: Weighs 1.5 grams or under Threshold: Weighs 2 grams or under
Stays firmly attached to the slide of the trombone.	Yes/No
Can be installed on and off the slide within 1 minute.	Yes/No
Will not break or deform after multiple uses.	Goal: Can withstand 10 lbs. of force Threshold: Can withstand 7 lbs. of force
Can accommodate different hand sizes.	Goal: Can fit finger sizes from 14.0 mm to 22.6 mm in circumference. Threshold: Can fit finger sizes from 15.2 mm to 20.2 mm in circumference.

c. Testing Methodology and Verification Plan

The goal of the verification plan was to make sure the trombone grip prototype met the specified design requirements through testing. Each test was made to evaluate the performance of the prototype, more specifically to test the material composition and functionality. The collected data was analyzed with the verification scorecard, so areas that needed improvement could be

identified.

Weight Test

- **Purpose:** Verify that the grip is **lightweight** to not be a burden on the musicians.
- **Measurement Method:** Weigh the prototype using a **high-precision scale**.
- **Equipment:** Digital scale with **±0.1 grams accuracy**.
- **Test Procedure:**
 1. Place the prototype on the scale and record the weight.
 2. Repeat the measurement **three times** to ensure precision.
 3. Compare the values against the target threshold.
- **Pass Criteria:** The grip must weigh **1.5 grams or under** to meet the goal and **must be ≤ 2 grams** to pass.

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Grip Stability Test

- **Purpose:** Ensure the grip **remains securely attached** to the trombone slide without moving.
- **Measurement Method:** Apply a **force** perpendicular to the grip and measure displacement.
- **Equipment:** Force gauge and caliper for displacement measurement.

Test Procedure:

1. Attach the grip to the trombone slide.
 2. Apply an increasing **lateral force on it** (1N, 2N, 3N, etc.).
 3. Measure the **change in position (mm)** after each force increment.
 4. Conduct the test **five times** for proper precision.
- **Pass Criteria:** The grip should show **no measurable movement** under normal handling conditions (≤ 0.1 mm displacement).

Installation Speed Test

- **Purpose:** Ensure that the grip can be **easily installed and removed within 1 minute.**
- **Measurement Method:** How long it takes for a user to install and remove the grip.
- **Equipment:** Stopwatch, user participants (at least 3 musicians). •

Test Procedure:

1. Provide a new user with the grip and instruct them to attach it.
2. Start the stopwatch when they begin and stop when the grip is fully installed.
3. Repeat for **three different users** to ensure repeatability.

- **Pass Criteria:** The grip should be **installed in ≤ 60 seconds.**

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Durability Test (Force Resistance)

- **Purpose:** Ensure the grip **does not break** after repeated use and can **withstand sufficient force.**
- **Measurement Method:** Apply increasing weight until the material bends or breaks.
- **Equipment:** Force gauge, weights (7-10 lbs.), testing stand. •

Test Procedure:

1. Secure the grip in a testing rig.
2. Apply a **gradually increasing load** starting from 5 lbs. up to 10 lbs.
3. Observe any signs of bending or structural failure.
4. Conduct **three trials** to confirm consistency.

- **Pass Criteria:** The grip should withstand at least **7 lbs. of force (threshold)** and ideally **10 lbs. or more.**

Adjustability (Hand Size Accommodation) Test

- **Purpose:** Ensure the grip fits a variety of **hand sizes** comfortably.
- **Measurement Method:** Measure the **inner circumference** of the grip at its smallest and largest adjustments.
- **Equipment:** Digital caliper for precise measurements.
- **Test Procedure:**
 1. Adjust the grip to its **smallest size** and measure the **inner circumference**.
 2. Adjust the grip to its **largest size** and measure again.
 3. Repeat the test **three times** to ensure precision.
- **Pass Criteria:** The grip must fit **finger sizes from 14.0 mm to 22.6 mm**. Add a penalty of -0.5 points for every 0.22 mm deviation.

d. Correlation Matrix and Verification Scorecard

The correlation matrix was used to evaluate how well the design requirements addressed the identified user needs. It allowed the team to determine which features, such as weight, durability, and adjustability, contributed most to user value. Based on this analysis, a weighted verification scorecard was created to assess the prototype’s performance. Each category reflected a key design goal and was scored during testing. The results highlighted both the strengths of the prototype and areas that needed improvement.

Table D.6: Correlation Matrix

	Light in weight	Does not slide around	Easy to install	Durable material	Accommodates different hand sizes	User Need Weight
Easier to hold instruments	9	3	1	1	3	5

Easier to transport instruments	0	3	0	3	0	4
Encouraging students to practice	9	3	9	3	3	0
Improving quality of instrument tone	1	0	0	0	0	1
Better material parts	3	1	1	0	0	3
Affordability	9	1	0	3	0	2
Importance ->	73	32	8	23	15	

Table D.7: Verification Scorecard Points Distribution

	Raw Percentage	Score Card Points (25)
Light in weight	48.3%	12
Does not slide around	21.2%	5
Easy to install	5.3%	1
Durable material	15.2%	4
Accommodates different hand sizes	9.9%	3
This sum should be 25! ----->		25

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Table D.8: Verification Scorecard

Requirement	Range	Score Rubric	Score
Light in weight	The weight of the grip is within 1-1.5 g.	Full points if the weight is within the range. -2 points for every 0.1 grams over and under the range.	12

Does not slide around	Should not move up and down the trombone during operation.	Full points if there is no change in position. -1 points for every 0.1 mm grip shifts.	5
Easy to install	Can be installed onto the trombone within 1 minute.	Full points if the grip can be taken on and off in 1 minute. -0.25 points for every 10 seconds over the range.	1
Durable material	7-10 lbs. can be placed on the top of the grip without bending or distortion.	Full points if the grip can withstand the weight. -1 point if the grip cannot withstand the weight.	4
Accommodates different hand sizes	Can fit finger sizes from 14.0-22.6 mm in circumference.	Full points if it can fit the size range. -0.5 points for every 0.2 mm less than the fitted size.	3

e. Prototype Preliminary Design and Mock-up



Figure 10.2. Preliminary Prototype 1

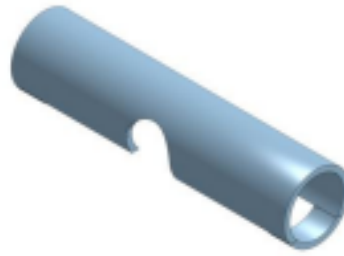


Figure 10.3. Preliminary CAD Mockup 1

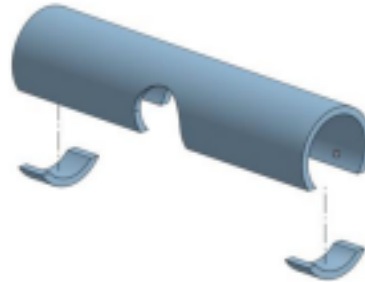


Figure 10.4. Updated CAD Model 1

The team first created a mock-up using paper and safety pins to get an idea of the basic shape of the ergonomic trombone grip. This initial prototype was less for functionality and more to visualize the basic shape. For the next version, the team considered materials like air-dry clay, paper clay, cardboard, and TPU filament to create a sturdier model. They also developed a 3D CAD model to better show the design and its proportions.

One major takeaway from the first prototype was the need for an internal component to keep the grip from rotating around the handle. Without this, users would have to constantly adjust their grip, which could cause strain. Fixing this issue became the next priority. In the updated CAD model, the team introduced a new idea—adding latches to the bottom of the grip that could be secured with hooks or magnets for better support.

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In the final model of the grip, the team planned to use TPU filament. This material was chosen because it is more flexible than PLA filament and less likely to break when attached to or removed from the trombone. The team decided on 3D printing since they only needed to complete the CAD model and print it, making the process much simpler. It also allowed for easier improvements to the prototype compared to using wood or metal.

Detailed Design Review

11. Prototype Fabrication and Evolution

The team began the design process by developing an initial concept focused on reducing wrist strain and improving comfort for trombone players during long practice and playing sessions. Research was conducted on existing products to identify gaps in the market and come up with a unique solution. After brainstorming different ideas, the team settled on a design that aimed to teach musicians the correct way to hold their instrument, solving a long-term problem instead of offering a temporary fix to pain like most other products. The initial concept was a grip that could attach onto the trombone slide, featuring holes for proper finger placement, that would not affect the instrument's original mechanics.

The first prototype was created using Onshape. It was designed as a semicylindrical grip that could snap onto the trombone slide, featuring holes for proper finger placement, without affecting the instrument's original mechanics (Figure 10.3). Once the 3D model was complete, the team created a physical prototype using paper and safety pins (Figure 10.2) to see the product's form in real life and look for improvements. Based on the feedback from the physical prototype, the 3D CAD model was updated to include an additional piece on the bottom that would ensure a firm grip and prevent slippage (Figure 10.4). After considering different manufacturing options, the team decided that 3D printing would be the best method for producing the model, as it was a complex shape that would be difficult to carve precisely from wood or metal. TPU filament was chosen for its flexibility and durability, making it ideal for repeatedly attaching and removing the grip from the trombone slide.

Before sending the final CAD file off, the firm attachment components were removed, as the pieces were so small they could easily be lost. The final model was then sent for 3D printing, and the team at this point was currently waiting for the printed prototype to be completed before beginning testing.

12. Detailed Design

a. Final Prototype Design

Much of the research indicated that trombone players experience pain due to improper instrument support. The left-hand bears most of the instrument's weight, and it is highly recommended by professionals that trombone players should use a hand grip for their left hand [11]. Specifically, the team's design aimed to teach trombone players how to hold their instrument correctly, so players could avoid

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the path to injury. When trombone players learned the proper left-hand form and support technique, they were less likely to experience back pain and could maintain steady airflow while playing [2].

b. Revised Grand Concept Design

Testing has not started, so we do not know any aspects of the design to improve upon yet.

13.Final Prototype Verification

The final prototype was designed based on the research conducted throughout the project. It reflected the key user needs identified during development, particularly the need for improved hand positioning and reduced physical strain. There were no non-standard components; the entire prototype consisted of a single integrated piece that was designed for 3D printing. This simplified the manufacturing process and ensured consistency in both form and function.

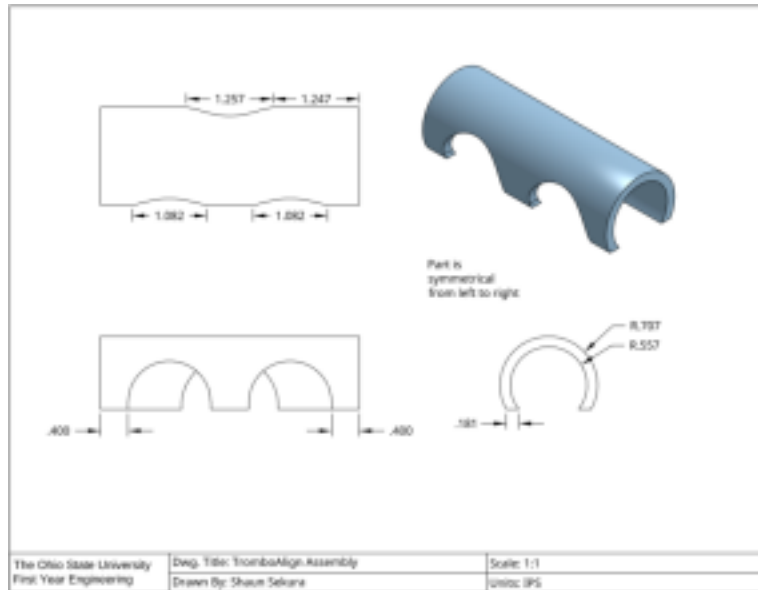


Figure C.1: Multiview Drawing of TromboAlign Clip

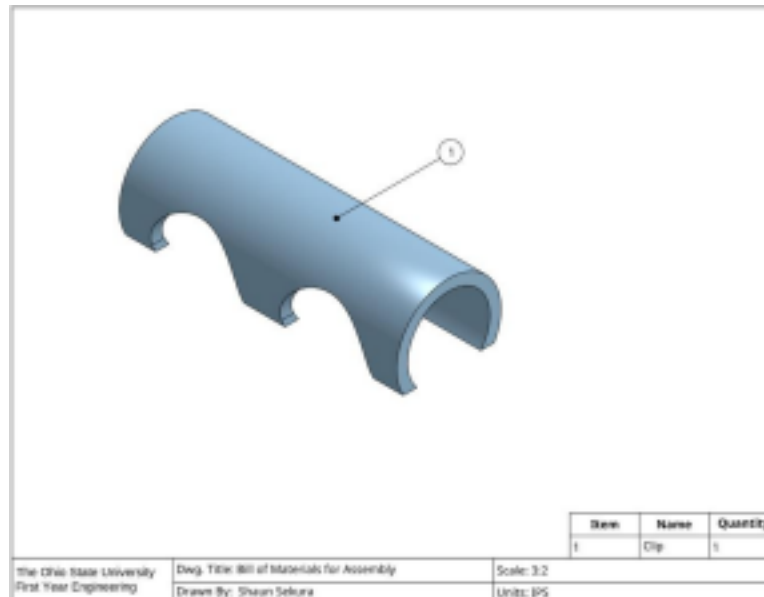


Figure C.2: Exploded Assembly Drawing of TromboAlign Clip

14. User Validation

The top three user needs were identified from the user needs chart and pairwise comparison chart. These needs included making the instrument easier to hold, making the instrument easier to transport, and reducing instrument-related costs. These priorities were based on common issues faced by trombone players during practice and performance. The most important problem was wrist strain from incorrect left-hand technique, and thus the team intended to design a product for correcting this technique.

The final prototype successfully solved this most important problem. The ergonomic grip included finger holes placed where the user should insert their left hand, correcting left hand form and reducing strain. It also indirectly improved airflow and playing stability as both issues are linked to overall form [22]. The grip was light so it would not be burdensome to the player. It was also small, which made transport easy. The grip also met affordability requirements. It cost around five dollars to produce one product which made it accessible for students, schools, and individual players.

To confirm the design was adequate, several tests were conducted. The Weight Test confirmed the team's claim that the grip was light. The Grip Stability Test verified the grip stayed securely attached and did not slide. The Installation Speed Test ensured installation and removal did not take more than 60 seconds for most users. The Durability Test confirmed that the grip could indeed withstand 10 pounds of force. Lastly, the Adjustability Test confirmed that the grip could accommodate a large range of hand sizes. The sum of these tests was what the team needed to confirm that design requirements were met.

The final product was useful for trombone players as it solved previously stated problems. With these problems solved, the team likely had an impact even on the lowest ranked user needs to improve practice time.

15.Value and Impact

Table D.9: Value Matrix

	Value Categories	
Stakeholders	Economic	Social
Professional Trombone Players	<i>A positive impact was that it reduced long-term healthcare costs by preventing injuries like tendonitis or carpal tunnel syndrome [9]. A negative impact would be the initial cost of purchasing the grip.</i>	<i>A positive impact is that it improved the player's physical comfort during practice, which enhanced the overall quality of life for the musician. A negative impact was that musicians may have been skeptical of using it.</i>
Music Stores	<i>A positive impact was that stores could have profited by selling the new product. A negative impact was that purchasing the stock and promoting the product was not free.</i>	<i>A positive impact was that a store could have showed their care for musician health by offering the product. A negative impact was that the grip could have been seen as just a gimmick by potential customers.</i>
Band Directors	<i>A positive impact was teachers saving time when showing students the proper grip for holding a trombone. A negative impact was that a music school may not have had the budget to purchase the product.</i>	<i>A positive impact was that students could learn the proper playing technique easily with the product. A negative impact is that some students could have been taught an incorrect method for using the grip.</i>
Parents of Student Trombone Players	<i>A positive impact was that parents could save medical costs on potential future student injuries. A negative impact was that the product is an additional expense for the student's music hobby.</i>	<i>A positive impact was that the parent could now see their child progress as a player. A negative impact was that some parents may have worried the product was a crutch or gimmick.</i>

The ergonomic trombone grip was intended to be self-sustaining in the long run but initially required start-up funding to support design, manufacturing, and product promotion. Maintenance and distribution were continual, ongoing costs. Revenue was generated through direct user fees and potential licensing opportunities in K–12 schools. These sources of income were expected to support continued production and allow the product to scale over time without ongoing subsidies. If users realized the product's

value, the team estimated that profit could reduce the need for subsidies.

The grip was designed to improve comfort for trombone players by correcting technique issues that cause repetitive wrist strain injuries. By correcting the student player's grip, students could now play for longer with better performance. Positive social results included the product improving player skill and thus allowing students to fit into more skilled groups. A negative impact was that some users were reluctant to adapt to the grip due to perceived crutch-like nature. There was also the possibility that students may have used the product incorrectly and thus not realized these benefits. Additionally, if the material was not durable enough, this could lead to frequent discarding of grips, which would likely decrease product trust rather than increase purchase frequency. To address

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these concerns, the team made sure the product was durable, intuitive enough for incorrect use to be unlikely, and effective at stopping player injury.

16. Project Recommendations and Next Steps

To move the TromboAlign product from prototype to production, several key resources are needed. These include access to flexible TPU filament, reliable 3D printing equipment, and testing tools to ensure continued refinement and quality control. Having a dedicated production workspace would also help streamline manufacturing efforts. In addition, CAD software will be essential for making further design adjustments based on user feedback, especially as the grip is adapted for broader hand size ranges and performance needs.

Beyond production, the next steps focus on bringing TromboAlign to market. This includes developing promotional materials, building partnerships with music stores, and launching an online store to reach individual customers. Support is also needed for packaging design and distribution logistics. Securing initial funding will be critical to cover early production costs and support outreach to schools, music programs, and private instructors. With these resources, TromboAlign has strong potential to become a widely adopted solution for improving comfort and technique among student trombone players.

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Appendix A. Evidence of Brainstorming

Brainstorming user groups

College students	Non-english speakers
Single mothers	People with ADHD
Gravelers workers	Terrible cat coffee drinkers

Musicians

- drums - guitar
- violin - trumpet
- flute - saxophone
- piano

Figure A.1: Brainstorming User Groups

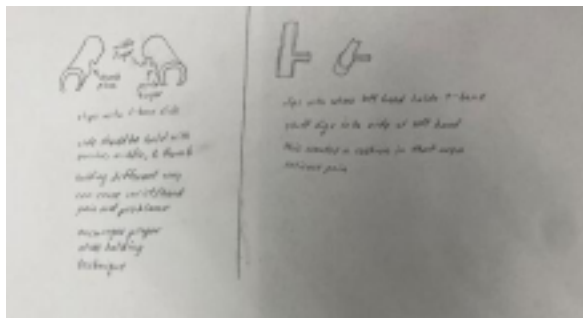


Figure A.2: Prototype Drawing 1



Figure A.2: Prototype Drawing 2



Figure A.3: Prototype Drawing 3

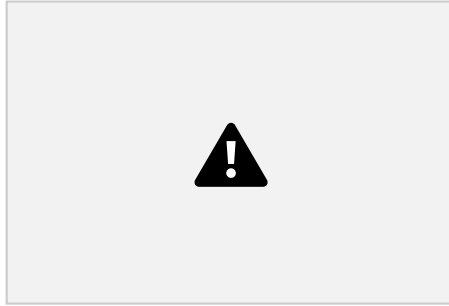


Figure A.4: Prototype Drawing 4

Appendix B. Research Methodologies

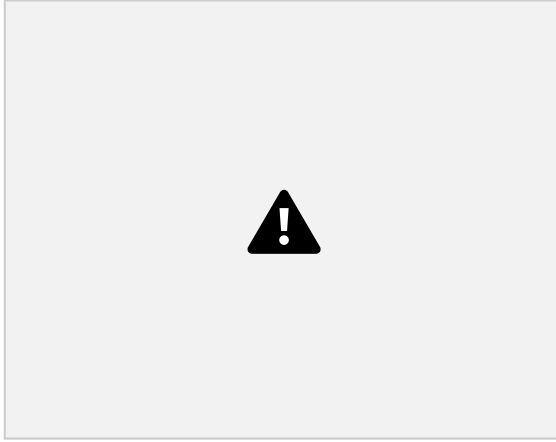


Figure B.1: Rough estimated calculations of trombone players in the US

Appendix C. Prototype Working Drawing Packet



Figure C.1: Multiview Drawing of TromboAlign Clip

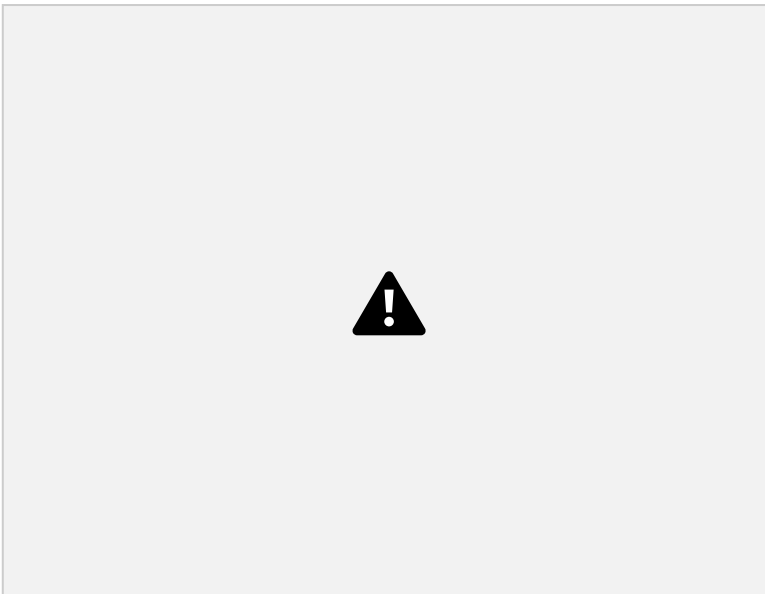


Figure C.2: Exploded Assembly Drawing of TromboAlign Clip

Appendix D. Additional Raw Data

Table D.1 Research Plan Table

Research Question	Quantitative Data	Qualitative Data
3.) Why do you want to play an instrument?	The Child as Musician: A Handbook of Musical Development [1]	Why Play – Trombone [2]
4.) What are the psychological benefits of playing the trombone?	A Prescription for Music Lessons [3]	The Psychological Benefits of Playing Brass Instruments: More Than Just Music [4]
3.) How does playing the trombone influence students' academic performance?	Investigating trainee music teachers' beliefs on musical abilities and learning: a quantitative study [5]	Relations among Motivation, Performance Achievement, and Music Experience Variables in Secondary Instrumental Music Students [6]
4.) How much time do trombone players typically practice in week?	If You Practice [7]	Trombone Fundamentals: How to Develop Your Own Daily Routine [8]
5.) What physical pain do trombone players have from playing?	Musculoskeletal and Psychological Assessments used in quantitatively based studies about musicians' health in brass players: A systematic literature review [9]	Play Without Pain: Repetitive Use Injuries and What to Do About Them [10]

Table D.2: Pairwise Comparison Chart

	Easier to hold instruments	Easier to transport instruments	Encouraging students to practice	Improving quality of instruments tone	Better material parts	Affordability	Total	Normalized	Predicted
Easier to hold instruments		1	1	1	1	1	5	5.0	5
Easier to transport instruments	0		1	1	1	1	4	4.0	4

Encouraging students to practice	0	0		0	0	0	0	1	1
Improving quality of instrument's tone	0	0	1		0	0	1	1.0	0
Better material parts	0	0	1	1		1	3	3.0	2
Affordability	0	0	1	1	0		2	2.0	3

Table D.3: Partial Competitive Matrix

User Need	Trombone SpineSaver™	Neotech™ Trombone Ultra-Grip	Calder pips Trombone Grip	ERGObass ERGObone	Edwards Bullet Brace
Easier to hold instruments					
Easier to transport instruments/ accessories					
Encouraging students to practice					
Improving quality of instrument's tone					
Better material parts					
Affordability					

Table D.4: Pugh Scoring Matrix

		Reference		Concept 1		Concept 2	
User Need	Weight (1-5)	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score
Easier to Hold the Instrument	5	5	25	5	25	5	25
Easier to transport accessory	4	3	12	3	12	5	20
Encourages students to practice	0	1	0	1	0	1	0
Improves quality of instrument's tone	1	1	1	1	1	1	1
Better Materials	3	1	3	1	3	5	15
Affordability	2	3	6	3	6	5	10
			0		0		0
			0		0		0
			0		0		0
			0		0		0
TOTAL			47		47		71

Table D.5: Design Requirement

Design Requirement:	Value(s):
Trombone grip should be lightweight.	Goal: Weighs 1.5 grams or under Threshold: Weighs 2 grams or under
Stays firmly attached to the slide of the trombone.	Yes/No
Can be installed on and off the slide within 1 minute.	Yes/No
Will not break after multiple uses.	Goal: Can withstand 10 lbs. of force Threshold: Can withstand 7 lbs. of force

Can accommodate different hand sizes.	Goal: Can fit finger sizes from 14.0 mm to 22.6 mm in circumference. Threshold: Can fit finger sizes from 15.2 mm to 20.2 mm in circumference.
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Table D.6: Correlation Matrix

	Light in weight	Does not slide around	Easy to install	Durable material	Accommodates different hand sizes	User Need Weight
Easier to hold instruments	9	3	1	1	3	5
Easier to transport instruments	0	3	0	3	0	4
Encouraging students to practice	9	3	9	3	3	0
Improving quality of instrument tone	1	0	0	0	0	1
Better material parts	3	1	1	0	0	3
Affordability	9	1	0	3	0	2
Importance ->	73	32	8	23	15	

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Table D.7: Verification Scorecard Points Distribution

	Raw Percentage	Score Card Points (25)
Light in weight	48.3%	12
Does not slide around	21.2%	5
Easy to install	5.3%	1
Durable material	15.2%	4
Accommodates different hand sizes	9.9%	3
This sum should be 25! ----->		25

Table D.8: Verification Scorecard Points Distribution

Requirement	Range	Score Rubric	Score
Light in weight	Weight of the grip is within 1-1.5 grams.	Full points if the weight is within the range. -2 points for every 0.1 grams over and under the range.	12
Does not slide around	Should not move up and down the trombone during operation.	Full points if there is no change in position. -1 points for every 0.1 mm the grip shifts.	5
Easy to install	Can be installed onto the trombone within 1 minute.	Full points if the grip can be taken on and off in 1 minute. -0.25 points for every 10 seconds over the range.	1
Durable	7-10 lbs. can be placed on the top of the grip without bending or distortion.	Full points if the grip can withstand the weight. -1 point if the grip cannot withstand the weight.	4
Accommodates different hand sizes	Can fit finger sizes from 14.0-22.6 mm in circumference.	Full points if it can fit the size range. -0.5 points for every 0.2 mm less than the fitted size.	3

Table D.9: Value Matrix

	Value Categories	
Stakeholders	Economic	Social

Professional Trombone Players	<i>A positive impact is that it will reduce long-term healthcare costs by preventing injuries like tendonitis or carpal tunnel syndrome. A negative impact is the initial cost of purchasing the grip.</i>	<i>A positive impact is that it improves the player's physical comfort when practicing, which can enhance the overall quality of life for the musician. A negative impact is that musicians may be skeptical of using it.</i>
Music Stores	<i>A positive impact is that profits could be boosted by selling a new product. A negative impact is that it will cost money to purchase stock and to promote the grip.</i>	<i>A positive impact is that promoting a health centered product can build their reputation as a customer-focused retailer. A negative impact is that the grip could be seen as just a gimmick by potential customers.</i>
Band Directors	<i>A positive impact is that it will save time when teaching students the proper grip for holding the trombone. A negative impact is that the school may not have the budget to purchase grips for all students.</i>	<i>A positive impact is that the grip will help teach students long-term healthy playing habits. A negative impact is that some students may incorrectly use the grip.</i>
Motivated Parents	<i>A positive impact is that it can help prevent future medical costs related to grip-related injuries. A negative impact is that it adds an extra expense on top of the cost of the instrument and lessons.</i>	<i>A positive impact is that it encourages children to continue playing the trombone by reducing physical discomfort. A negative impact is that some parents may worry their child will become too dependent on the trombone grip.</i>

Appendix E. Project Management Schedule and Meeting Minutes

17. Project Schedule

a. PERT Schedule

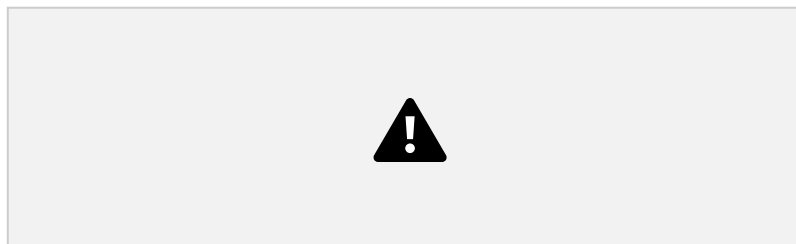


Figure E.1: PERT Schedule

The team planned to complete the final CAD model of the prototype and order the TPU filament from the ED store by April 1st. The team planned that once both tasks were finished, the first prototype would be 3D printed, with a goal to complete it by April 5th. Afterward, the prototype was to go through the

verification plan and scorecard process by April 10th. Based on the results, any necessary improvements were to be identified by April 15th.

Cathy is responsible for all 3D printing tasks. Shaun will handle the 3D CAD model. Nathan and Jean will work together on testing the prototype.

b. Gantt Chart



Figure E.2: Gantt Chart

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18. Prototyping Workday 1 Meeting Minutes

3/28/2025 | 12:40-1:35pm | In person | Cathy, Shaun, Jean, and Nathan

Objective Statement

The purpose of this meeting was to allocate tasks among team members in creating a second design of the product. The team also determined which parts to order from the EED store to complete the design.

Completed Tasks

- Worked on ways to systematically test, take measurements, and collect data on our prototype to improve upon the design (**Jean**)
- Worked on 7 different design requirements for our final prototype design (**Shaun**) - Used available materials to build a preliminary mock-up for our prototype (**Nathan**) - Used Cura to determine the amount of TPU filament needed and the printing time (**Cathy**)

Tasks to be completed

- Ordered the TPU filament from the EED store (**Cathy**)
- Refined the design so it fits better on the trombone slide (**Shaun**)

- Tested the prototype for problems (**Jean, Nathan**)

Project Timeline



Figure E.3: Workday 1 Gantt Chart

Decisions

The team decided to use TPU filament to create the next prototype of the design. This created a better prototype because it was more durable, especially during testing. After the intensive testing, the team decided to refine the design so the product could handle being attached to the trombone slide for an extended period.

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19. Prototyping Workday 2 Meeting Minutes

4/9/2025 | 11:10 am – 12:30 pm | In person | Cathy, Shaun, Jean, and Nathan

Objective Statement

The purpose of this meeting was to work out the details of the project and fine tune the team's report before the completion of the prototype.

Completed Tasks

- Worked on the fixing formatting errors for assignment DD3 (**Cathy**)
- Created an exploded assembly, bill of materials, and detailed drawings (**Shaun**)

Tasks to be completed

- Tested the prototype for problems (**Jean, Nathan**)

Project Timeline

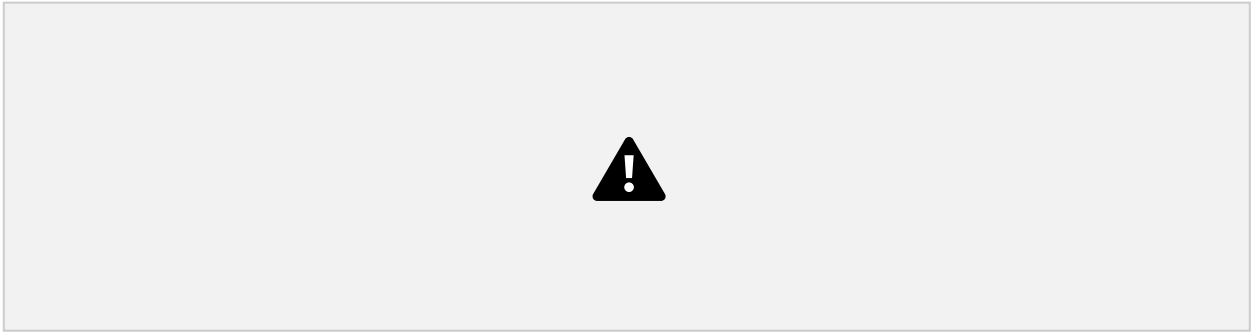


Figure E.4: Workday 2 Gantt Chart

Decisions

The team was unable to move on at this point as the prototype was still in the manufacturing process.

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20. Prototyping Workday 3 Meeting Minutes

4/11/2025 | 12:40 pm – 1:35 pm | In person | Shaun, Jean, and Nathan

Objective Statement

The purpose of this meeting was to work out the details and fine tune the team's report prior to the completion of the prototype.

Completed Tasks

- Wrote up the meeting minutes for today (**Jean, Nathan, Shaun**)

Tasks to be completed

- Fixed the prototype so it could attach to the trombone slide properly (**Shaun**)
- Tested the prototype for problems (**Jean, Nathan**)

Project Timeline



Figure E.5: Workday 3 Gantt Chart

Decisions

The team had determined the exact issues of the prototype and planned to have a refined prototype built by the next week.

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21. Prototyping Workday 4 Meeting Minutes

4/15/2025 | 12:40 pm – 1:35 pm | In person | Cathy, Shaun, Jean, and Nathan

Objective Statement

The purpose of this meeting was to revise the working document and improve the prototype up until the project's completion.

Completed Tasks

- Wrote up the meeting minutes for today (**Cathy, Jean, Nathan, Shaun**)

Tasks to be completed

- Improved prototype's ability to attach to trombone slide (**Cathy, Shaun**)
- Tested the prototype for problems (**Jean, Nathan**)
- Revised the working document (**Cathy, Shaun, Jean, Nathan**)

Project Timeline



Figure E.6: Workday 4 Gantt Chart

Decisions

The team determined the issue with the initial prototype and planned to have a refined prototype by the end of this week. The team was also continuing to revise the working document and test the prototype for other possible issues.

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Appendix F. Team Working Agreement

Team Working Agreement

Spring 2025

3/4/2025. Revised Date

1. Team Information ENGR 1182.02 Section 8078

Table A

Christopher Ratcliff and Adam Kling

Contact Information:

Name	Email Address	Phone Number
Shaun Sekura	Sekura.2@osu.edu	440-319-5672
Cathy Kuo		
Jean Richard Fenelon		
Nathan Muzumdar		

2. Team Values & Goal

The team's top values were perseverance, discipline, resilience, teamwork, and inclusivity. The team's expectation of timeliness was to complete the work at least 1-2 days before the actual due date, and the goal for the assignment was for it to be completed accurately, with correct formatting, with specificity, and cohesively. Another goal of the team was splitting work inclusively, including everyone in the project and recognizing the contributions each member brought to the group.

3. Communication and Meetings

The team has a group chat that we utilize as our primary method of communication. The expected response time should be around a day, unless there are external circumstances. The team currently plans to meet weekly to complete our assignments, although this is subject to change during the semester, as coursework and other priorities can become heavy as the semester progresses. The location of our primary meetings is not set in stone, but most meetings are planned to be held directly after the engineering classes on Tuesday and Friday.

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4. General Expectations and Group Norms

The team should work together and put their appropriate effort into every single assignment. Team members should show up to every single class and lab (except for extenuating circumstances including sickness, personal emergency, etc). If a teammate misses a class, they should message other teammates as soon as possible notifying them of the absence. During class, team members should be present and participate, including contributing to projects. In between classes, work will be divided between members, or we will find a meeting time to come together and work on an assignment together. All of us will make sure we have enough time to finish a project before the deadline. Documents will be shared with each other through OneDrive. Every member should have their work completed a day before the deadline. If a member is struggling, they should text the group chat for help as soon as possible.

5. Individual Team Member Responsibilities

Assignments will be divided based on what each team member can contribute. Every member of the team will have to complete their fair share of each assignment. We will split the assigned parts beforehand to make sure everyone is involved. Everyone will rotate roles in each assignment, but everyone should have the responsibility of checking over everyone else's work.

6. Conflict Resolution:

Accountability:

Members of the team will hold each other accountable by providing progress reports and frequent check-ins during meetings. Every team member is responsible for completing work accordingly to meet project deadlines. If this is not followed, the issue will be discussed in a team meeting if a member is continuously unable to meet expectations.

Dealing with non-participation:

The entire group will first have a polite, private discussion with the person if they are not contributing properly to identify any underlying problems. The team will involve the professor or GTA for additional assistance if there is no improvement in team members' performance.

Addressing Inappropriate Interaction:

A confidential conversation with the concerned team member will be held right away to address inappropriate interactions. Effective and respectful communication will be emphasized by the team. If the student exhibits unacceptable behavior repeatedly, the professor or GTA will be called in to help handle the issue.

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Redefining objectives, duties, and expectations:

If the entire team decides that adjustments are required because of new ideas, changes in the project's scope, or team dynamics, then it is permissible to revise goals, expectations, and duties, and all team members will be informed of the revisions, and such revisions will be properly documented.

Managing Concerns About Inclusivity:

To address inclusivity issues, the team will have an open discussion if one person is not operating in an inclusive manner. If necessary, the group will provide resources or instruction related to diversity and inclusivity. For additional action, persistent problems will be brought to the attention of the instructor or GTA.

Instructor, GTA, or UTA involvement:

If there is a violation of the team agreement that requires outside help, or if conflicts cannot be handled internally, the team will involve UTAs, GTAs, or the instructor. If a team member continuously misses deadlines and does not attempt to figure out how they can solve the issue, the team will involve the instructor, GTA or UTAs. When interpersonal disputes remain unsolved and interfere with team development and productivity or a team member behaves in a way that is offensive, discriminatory, or detrimental to the team's inclusive culture UTAs, GTAs, or the instructor will be called for assistance.

7. Expectations of Faculty and GTAs:

The team will still turn in a finished project on the due date, but if a team member doesn't follow through on this agreement, the staff may be informed. To assist the team in resolving disputes and obstacles, faculty and GTAs are expected to offer timely feedback and assistance. The faculty and GTAs will lead the group in identifying helpful solutions. Additionally, they will keep an eye on the team's development and make sure the working agreement is being followed.

8. Team Signatures



Name: Cathy Kuo

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Name: Shaun Sekura



Name: Jean Richard Fenelon



Name: Nathan Muzumdar